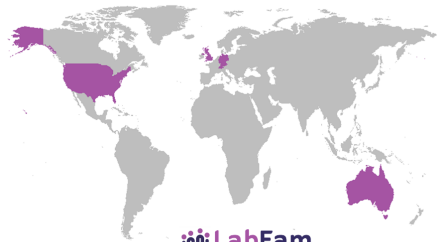


LabFam Individual Biographies (LIB)

Harmonised Family and Employment Histories Based on Panel Surveys
Workshop

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 **LabFam**
Individual Biographies

Outline

Why LIB? Motivating the Harmonization

Source Data: Five Panels, 1968–2024

Constructing Fertility Biographies

Constructing Partnership Biographies

Constructing Employment Biographies

Technical Validation

Compatibility with CNEF/CPF

Working with the LIB Codebase

Running LIB

Getting the data

Running the code

Materials

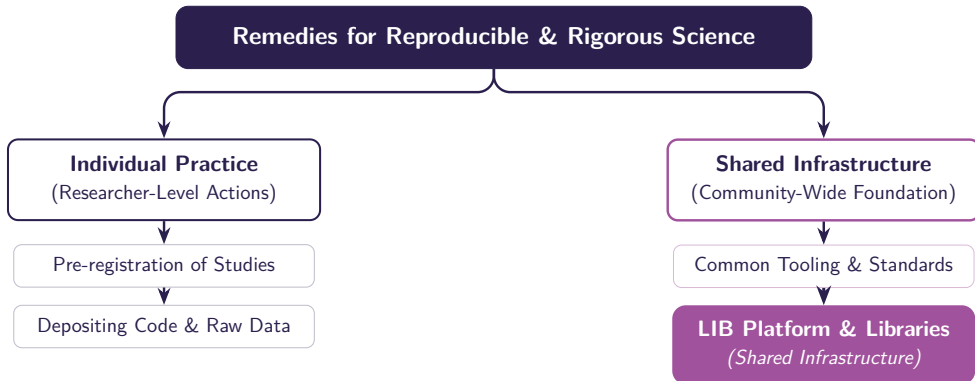
References

Social Science Has a Reproducibility Problem

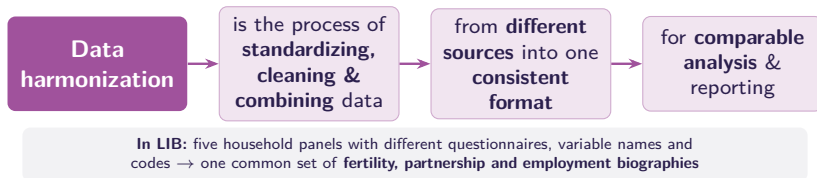
- ▶ Researchers repeatedly struggle to reproduce each other's results from the same databases, which Freese, Rauf, & Voelkel (2022) argue undermines the credibility of published findings.
- ▶ Eubank (2016) notes that without reproducible workflows, researchers cannot easily build on one another's work – everyone pays the setup cost again.
- ▶ In a field experiment, Huntington-Klein et al. (2021) had fellow economists attempt to replicate a published study: some obtained opposite results, and some could not even reconstruct the original sample.
- ▶ Even for a single, widely used panel, replication code is rarely shared: Fink & Marcus (2025) show that among SOEP-based publications, code availability has improved over time but remains low and varies strongly across fields.

Two Routes to Reproducible Research

Two main remedies exist to improve scientific reproducibility: **individual practices** (e.g., pre-registration) and **shared infrastructure**. **LIB belongs to the second category.**



LIB's Route: Harmonizing Data as Shared Infrastructure



Benefits

Cross-country comparison

same definitions and codes in every country

Better data quality

inconsistencies found and fixed along the way

Easier sharing

one documented format others can reuse

Faster research

analysis starts from ready-made biographies

Reproducible workflow

open code makes every step transparent

The idea of LIB

Comprehensive

Utilises panel, calendar and retrospective questionnaires to reconstruct overlapping monthly histories

Open

Provides an open-source code, lowers the entry barriers

Flexible

Allows choosing time span, set of countries and dimension of individual biographies, any further modification

Inclusive

Part of a larger endeavour, allows adding other surveys

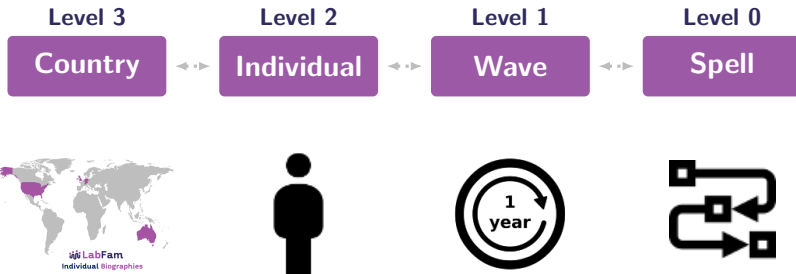
Compatible

CNEF (Frick et al., 2007); CPF (Turek et al., 2021); P&F GGS (Perelli-Harris et al., 2015)

From Messy Panels to Comparable Histories

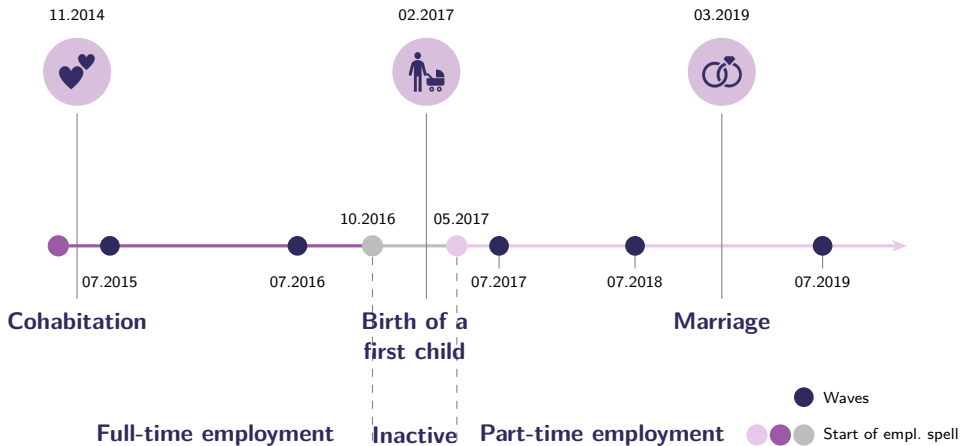
- ▶ LIB harmonises individual-level longitudinal data into continuous life-course trajectories for people observed in five major household panels.
 - ▶ Germany SOEP (German Socio-Economic Panel)
 - ▶ Switzerland SHP (Swiss Household Panel)
 - ▶ United Kingdom BHPS / UKHLS (British Household Panel Survey Understanding Society)
 - ▶ United States PSID (Panel Study of Income Dynamics)
 - ▶ Australia HILDA (Household, Income and Labour Dynamics in Australia)
- ▶ Instead of a **status recorded at each interview**, LIB reconstructs **spells: continuous episodes with a defined start and end, during which a status stays constant**.
- ▶ Data become timelines of episodes, not point-in-time snapshots – enabling precise measurement of durations and transitions across the whole life course.

Structure of Data



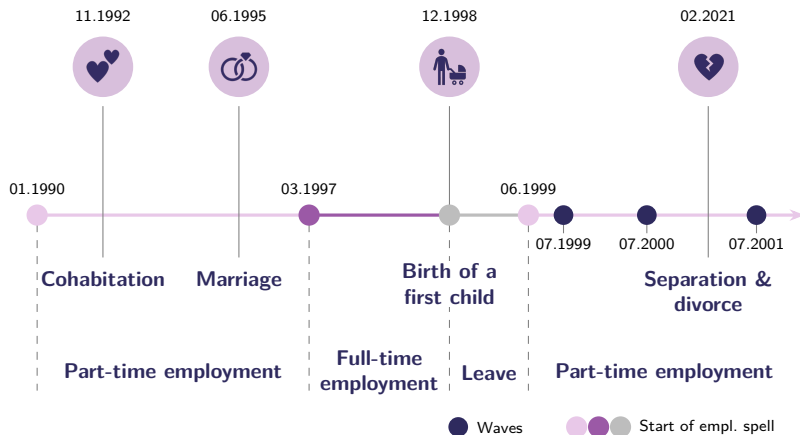
Data are organised in four levels: **country** → **individual** → **wave** (detailed annual characteristics, e.g. work conditions) → **spell** (episodes with exact start and end dates and duration, e.g. of unemployment). Spells are not nested within waves: one episode may last for years and span several waves, while another may be shorter than a year, so several spells can occur between two subsequent waves.

Scheme 1



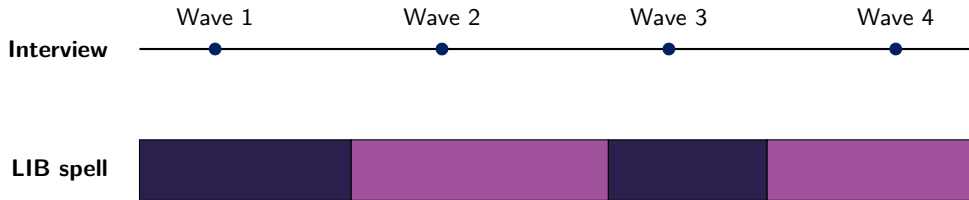
Example 1: a woman in her 30s, interviewed in five waves (2015–2019). At every interview she was employed, full- or part-time. Between the 2nd and 3rd wave, however, she was **inactive after the birth of her first child**: an episode that wave-based data alone would miss.

Scheme 2



Example 2: this person enters the panel only in 1999 and is interviewed in three waves. Yet LIB also recovers what happened **before the first interview**: a decade of employment history, cohabitation, marriage, and the birth of a first child followed by a leave. The separation and divorce fall **between two waves**, so their timing, too, comes from the event histories rather than from the wave snapshots.

Spells vs. Snapshots



A survey interviews you once a year; your job, marriage, or child does not wait for the interview. LIB fills the gaps between waves using calendar modules and retrospective questions, so a mid-year transition shows up even though nobody asked about it that month.

Why Spells, Not Snapshots: Example Studies

Monthly impact functions

*Hudde & Jacob (2023),
Sociological Science*

Most panel studies model change around life events in yearly steps. Using the month of each event, the authors trace outcomes before and after major life events far more precisely.

Lesson: the information is already in the data; yearly snapshots throw it away.

Union status around first birth

*Perelli-Harris et al. (2012),
Population Studies*

Union and fertility histories for eleven European countries show whether cohabiting couples marry before conception, during pregnancy, after the birth, or separate.

Lesson: the *order* of events can only be seen with dated histories.

Panel vs. life-course data

*Manzoni, Luijkx & Muffels (2011),
Quality & Quantity*

Compares labour market transitions in West Germany measured with panel data and with life-course (retrospective history) data, and explains why the two sources give different pictures.

Lesson: how transitions are measured shapes what we find.

Month-level information is routinely collected but rarely used: most studies still compare yearly waves (Hudde & Jacob 2023; see also Watson 2009).

How we store data? - What LIB Delivers: Three Domains

Fertility

Number and timing of biological (and adopted) children.

pid	KID_1	KID_Y1	KID_2	KID_Y2
101	1	2012	1	2016
102	1	2018	0	NA
103	0	NA	0	NA

Available Variables

- ▶ **KID_\$**: Child indicator (1/0)
- ▶ **KID_MS**: Month of birth
- ▶ **KID_SS**: Biological sex
- ▶ **NR_KIDS**: Total kids

Partnership

Union formation, cohabitation, marriage, separation.

pid	STATUS_1	START_Y_1	END_Y_1
101	Cohabit	2010	2015
102	Marriage	2018	NA
103	Single	2020	2022

Available Variables

- ▶ **START_MS**, **END_MS**
- ▶ **PARTNER_IDS**
- ▶ **END_SINGLE/UNIONS**
- ▶ **DIVORCES/Y/M**

Employment

Monthly labor-market status and job spell boundaries.

pid	STATUS_1	ENTRY_Y1	EXIT_Y1
101	Work	2008	2014
102	NILF	2015	2018
103	Work	2019	NA

Available Variables

- ▶ **ENTRY_M/D\$**: Entry date
- ▶ **EXIT_M/D\$**: Exit date
- ▶ **STATUS**: 1=Work, 2=Unemp, 3=NILF

Each domain is stored the same way: **one row per person ('pid')**, ordered chronologically by spell ('\$ = 1, 2, ...').

Standing on Prior Work



Frick et al. (2007)

Cross-National Equivalent File. Harmonizes main-survey variables at interview date.

pid	wave	d11107
101	1	0
101	2	1
102	1	2

Key Features

- ▶ **Type:** Harmonized dataset
- ▶ **Domains:** Income, health, labor, education, demography
- ▶ **Granularity:** Point-in-time yearly panel



Turek, Kalmijn, & Leopold (2021)

Comparative Panel File. Builds on same panels, centered on main-wave variables.

pid	wave	nkids
101	1	0
101	2	1
102	1	2

Key Features

- ▶ **Type:** Open-source code pipeline
- ▶ **Domains:** Work, family, health, education, well-being
- ▶ **Granularity:** Annual panel waves

Harmonized Histories

Perelli-Harris et al. (2015)

GGs retrospective histories. Non-linear life course trajectories across sources.

pid	kid_1	kid_y1	kid_y2
101	1	2012	NA
102	1	2010	2014
103	0	NA	NA

Key Features

- ▶ **Type:** Standardized spell dataset
- ▶ **Domains:** Fertility & partnership histories
- ▶ **Granularity:** Spell-based sequences

Like **CPF**, **LIB** provides open code users can run and modify (unlike static datasets like CNEF). However, LIB differs from both wave-based approaches by reconstructing continuous monthly spells **capturing transitions that occur between interview waves.**

How LIB Extends These Approaches

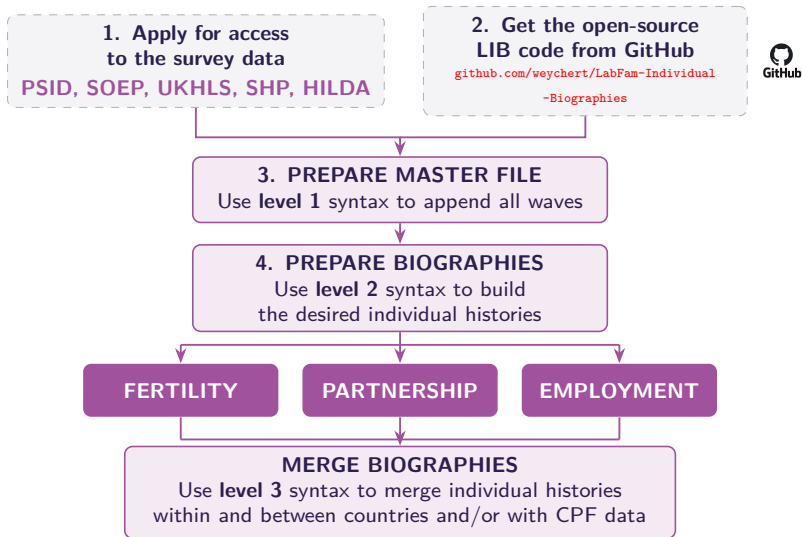
- ▶ Beyond CNEF/CPF, LIB adds (1) **calendar data** to bridge employment histories between main waves, and (2) **retrospective components** to reconstruct histories before survey entry.
- ▶ Unlike the Harmonized Histories, LIB also covers **employment**, not just fertility and partnership – the GGS retrospective data cannot support this.
- ▶ LIB is not purely retrospective: it leans heavily on long-running **panel** data, reducing recall and survival bias relative to a fully retrospective design.
- ▶ Caveat: comparisons with the Harmonized Histories should be made cautiously, given that source's shorter panel structure and thinner pre-interview employment content.
- ▶ Many of the questions LIB is designed for – e.g. how employment uncertainty shapes fertility and partnership behaviour – have so far been studied largely **within single countries** (Drobni 2002; Kreyenfeld 2010; Di Nallo & Köksal 2023; Osiewalska, Matysiak, & Kurowska 2024), since preparing comparable data for several countries at once was previously too costly.

Two Components of LIB

1. **Source data** – the original longitudinal datasets. Users must obtain these independently from each panel's data provider; LIB does not redistribute them.
2. **Open-source code** – written in R (version 4.4.0), which reconstructs LIB histories once access to the source data is granted.

The code is modular: outputs can be tailored by country, period, or life-course domain, and every transformation stays visible and inspectable.

Work Flow



Panel Characteristics

Survey	Country	Frequency	Sample size (approx.)	Time span
PSID	USA	Annual to 1997, biennial since 2003	18,000 ind. / 5,000 hh	1968–2021
SOEP	Germany	Annual	30,000 ind. / 20,000 hh	1984–2021
BHPS/UKHLS	UK	Annual	13,000 ind. / 7,000 hh	1991–2023
SHP	Switzerland	Annual	7,000 ind. / 5,000 hh	1999–2023
HILDA	Australia	Annual	17,000 ind. / 12,000 hh	2001–2020

Version numbers at time of writing: PSID 44, SOEP 40, BHPS 18/UKHLS 14, SHP 24, HILDA 20.

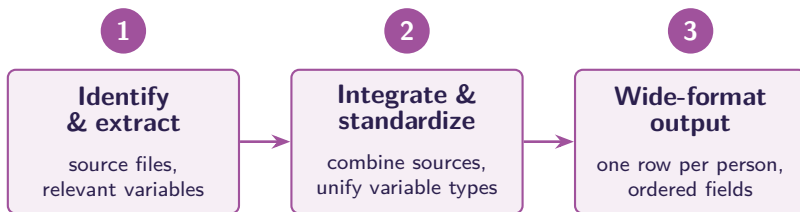
Access & Governance Notes

- ▶ Each database must be obtained **independently from its original provider**, under that provider's own data-access agreement.
- ▶ **LIB shares code, not data**: the harmonisation scripts are open on [GitHub](#), but the harmonised biographies cannot be redistributed, because the source data are licensed.
- ▶ LIB used the most recent panel releases available under its own agreements; newer releases may exist, since all five surveys are ongoing.
- ▶ Taken together, the five panels span **1968 to 2024**.
- ▶ Access conditions and timelines vary widely: some panels are available the same day, others need a signed contract or an application. **Start access applications well in advance.**

Where and How to Get the Data

Panel	Provider (link)	How to access	Cost	Waiting time
HILDA	DSS, via Australian Data Archive	Online request plus signed Confidentiality Deed Poll; an Australian collaborator is needed . Users abroad get the General Release only.	Free	Weeks (request reviewed)
SOEP	DIW Berlin, RDC SOEP	Data distribution contract signed by your institution and DIW, then encrypted download.	Free	Weeks (contract signed by both sides)
SHP	FORS, via SWISSUbase	SWITCH edu-ID login and online usage contract, then direct download.	Free	Same day
BHPS/ UKHLS	UK Data Service	End User Licence (SN 6614): download after registration, but no month of birth. Special Licence (SN 6931): application with justification.	Free	EUL: same day SL: weeks
PSID	ISR, University of Michigan	Registration for public data via the Data Center; restricted data only through a contract.	Free	Public data: same day

Three Steps, Applied to Every Panel and Domain



This same three-step logic is repeated independently for **fertility**, **partnership**, and **employment**, in each of the five panels.

Fertility Biography Data: Content and Cross-Panel Variation

What the Fertility Biography Records

- ▶ Date of birth of each biological or adopted child (month and year)
- ▶ Child's biological sex
- ▶ Birth order (parity), both during and prior to survey participation

Data provision: Some providers distribute a dedicated file; others require combining main-survey and retrospective files.

Cross-Panel Variation in Fertility Data

- ▶ **Coverage:** Resident-only vs. resident *and* non-resident children (resident-only biases toward recent births).
- ▶ **Type:** Whether adopted children are distinguished from biological children.
- ▶ **Precision:** Birth dates recorded to the month (most common) or only to the year.

Fertility Biography: SOEP and PSID - distributed dedicated file

SOEP Fertility Biography

- ▶ Based on `biobirth.dta`, distributed by DIW and prepared by Schmitt (2020).
- ▶ Draws on the main questionnaire plus a retrospective survey completed at first participation.
- ▶ Monthly precision on birth timing, with clear parent–child linkages.
- ▶ Lists *all* biological children, regardless of residence status.

PSID Fertility Biography

- ▶ Built from the **1985–2021 Childbirth and Adoption History File**, covering all eligible individuals within a PSID family.
- ▶ Provides month/year of birth, sex, and parity for both biological and adopted children – but does not distinguish resident from non-resident children.
- ▶ Retrospective reports collected in 1985 reconstruct complete birth histories, available only for individuals still in the study that year.

Fertility Biography: BHPS/UKHLS and SHP

BHPS/UKHLS Fertility Biography

- ▶ **BHPS:** Non-resident children (`b_childnt`, `k_childnt`, `l_childnt`) from waves 1, 11, and 12 (sex and birth month/year).
- ▶ **UKHLS resident children:** `xhhrel.dta` (family matrix) gives child IDs, merged with `xwavedat.dta` for sex and birth date (1991–2022).
- ▶ **UKHLS non-resident children:** `a_natchild` and `f_natchild` (waves 1 and 6) give sex and birth year.
- ▶ Combined and de-duplicated by individual ID, child parity, and birth year.

SHP Fertility Biography

- ▶ **Resident children:** Parity-level data from `shp_mp.dta`, linked via `idmoth_/idfath_` for month, year, sex.
- ▶ **Non-resident children:** Sex and birth year from `shpWave_p_user.dta` (waves 1–15 only; stopped in wave 14, so post-2013 joiners lack this data).
- ▶ **Retrospective supplement:** 2013 fertility module (`shpii_fa_user.dta`) adds birth year for all biological children, including non-survivors.

HILDA Fertility Biography: Construction Method

Data Sources & Linkages

- ▶ **Non-resident children:** Processed wave-by-wave from `ncage1-13` (age) and `ncsex1-13` (sex) in the combined file.
- ▶ **Resident children:** Linked via biological parent IDs (`hhbmxid/hhbfxid`) and household ID (`hhrid`) within wave.
- ▶ **Birth year estimation:** Non-resident age is subtracted from interview wave year ($\text{Year} - \text{Age}$).

Aggregation & Post-Processing

- ▶ **Parity recalculation:** Resident and non-resident records are merged into long format and ordered by birth year to assign child parity.
- ▶ **Longitudinal consolidation:** Transformed to wide format per wave; respondents' *most recent* available wave info is retained.
- ▶ **Zero-parity inclusion:** Appends childless respondents from master file to construct complete cross-sectional history.

Partnership Biography Data: Content and Cross-Panel Variation

What the Partnership Biography Records

- ▶ Timing of each union formation (cohabitation or marriage)
- ▶ Union end date and reason: separation, divorce, or partner's death
- ▶ The date cohabitation converted into marriage, when applicable

Data provision: Wide format, one row per individual from age 15 onward, with spells ordered chronologically.

Cross-Panel Variation in Partnership Data

- ▶ **Coverage:** Co-resident partners vs. non-coresident partnerships (LAT – Living Apart Together).
- ▶ **Type:** Distinguishing direct marriage, cohabitation prior to marriage, and informal cohabitation.
- ▶ **Precision:** Start/end dates recorded to the month (the common case) vs. year only.

Partnership Biography: BHPS/UKHLS and PSID - distributed dedicated file

BHPS/UKHLS Partnership Biography

- ▶ **Source:** *Marital and Cohabitation Histories, 1991–2022* (phistory_long.dta, licence 8473), covering pre-survey periods.
- ▶ Contains cohabitation/marriage spells, partner IDs, and start/end dates.
- ▶ Cohabitation transitioning to marriage is recorded as **two** separate spells.
- ▶ Adds singlehood periods and flags for union dissolution reasons.

PSID Partnership Biography

- ▶ **Source:** The **1985–2021 Marriage History File**.
- ▶ **Key Limitation:** PSID records marriages only; cohabitation is not tracked in this file or the main survey, making it absent from the biography.

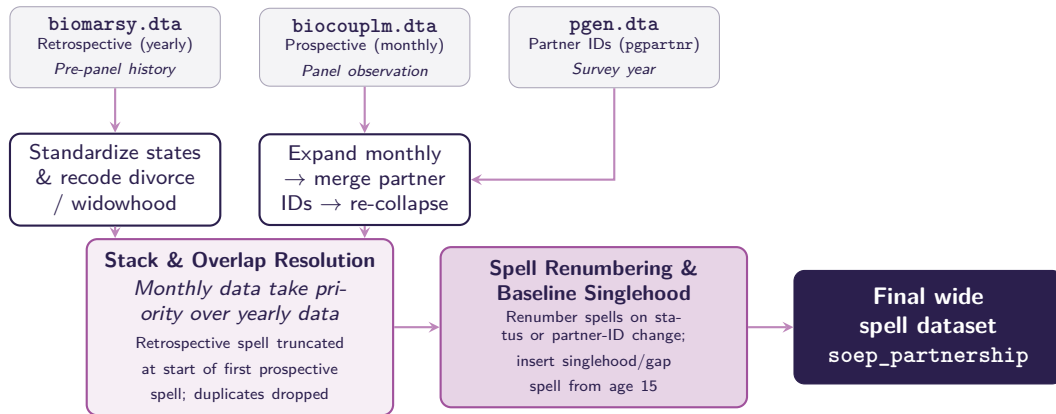
Data Integration & Resolution

- ▶ **Source prioritization:** Retrospective annual data (`biomarsy`) are stacked first, followed by prospective monthly data (`biocouplm`)[cite: 2].
- ▶ **Censoring & Imputation:** If a retrospective spell is right-censored at panel entry, its end year is imputed from the start date of the overlapping prospective spell[cite: 2].
- ▶ **Duplicate removal:** Prospective monthly data take precedence; redundant overlapping retrospective spells are dropped.

Monthly Expansion & Linkages

- ▶ **Partner ID linking:** Prospective spells are expanded into monthly person-period observations to merge partner IDs (`pgpartner`) from `pgen` by month/year[cite: 2].
- ▶ **Spell re-construction:** Data are collapsed back into distinct spells, incrementing spell numbers whenever status or partner ID changes[cite: 2].
- ▶ **Early singlehood:** Adds a baseline "gap/single" spell from age 15 up to the start of the first reported partnership.

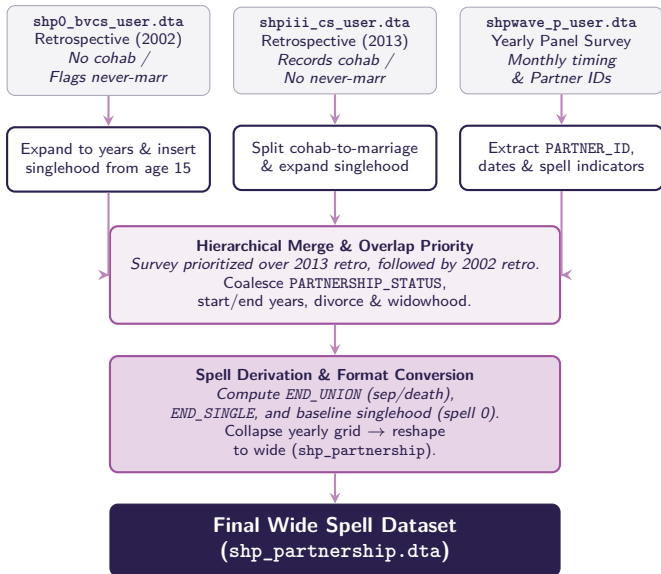
SOEP Partnership Biography: Pipeline



SHP Partnership Biography

- ▶ Built from the annual survey (`shpwave_p_user.dta`) plus two retrospective modules: 2002 (`shp0_bvcs_user.dta`) and 2013 (`shpiii_cs_user.dta`).
- ▶ Retrospective modules report only start/end *years*, no partner IDs
- ▶ 2002 module: no cohabitation, but flags never-married respondents. 2013 module: records cohabitation, but no never-married flag.
- ▶ Main survey gives monthly timing and partner IDs – prioritized on overlap, as it is closest to interview date.

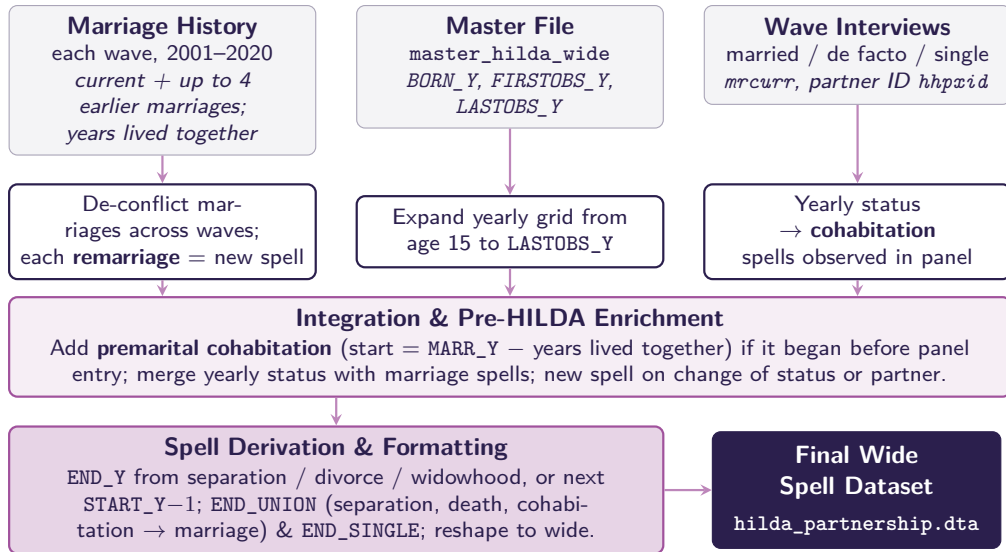
SHP Partnership Biography: Pipeline



HILDA Partnership Biography

- ▶ Built from the `letter_Combined.dta` files: current/most recent union plus up to four prior unions.
- ▶ Precise month/year start and end dates are available **only for the most recent union**.
- ▶ Ongoing cohabitation at interview is captured, including partner ID.
- ▶ **Issue:** no full pre-entry cohabitation history, so coverage before panel entry is less complete than in other panels.

HILDA Partnership Biography: Pipeline



Employment Biography Data: Content and Cross-Panel Variation

Why Employment Is the Hardest Domain?

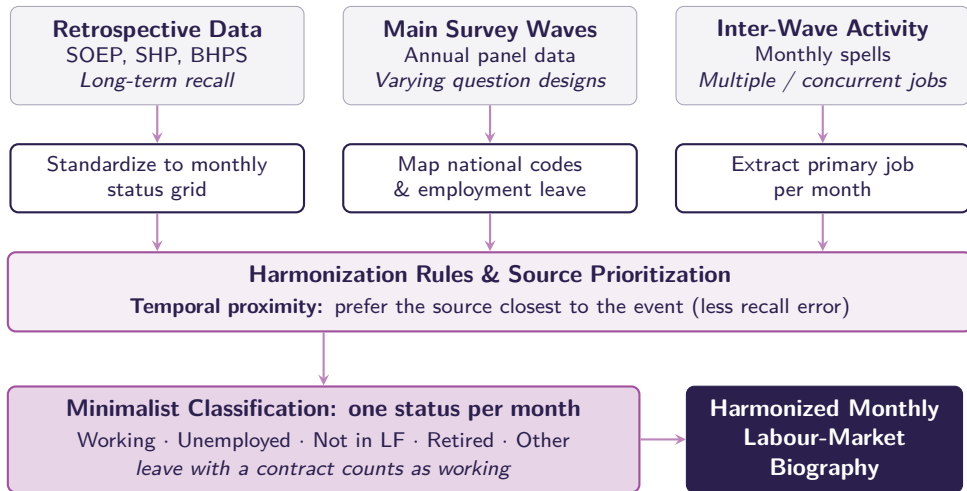
- ▶ People can hold multiple jobs at once, or combine work with other activities such as education.
- ▶ Panels differ in: information collected between waves, availability of retrospective data (SOEP, SHP, BHPS only), main-survey question design, and even occupation/industry coding schemes.
- ▶ LIB therefore takes a deliberately minimalist approach: one main labor-market status per month.

Status categories: working, unemployed, not in the labor force, retired, other. (Someone on leave but still under contract, e.g. maternity leave, counts as employed; all contract types count.)

Two core rules:

1. When sources overlap, prioritize the source **closest in time** to the spell – this minimizes recall error (Watson 2009).
2. When several statuses apply in one month, resolve via a hierarchy (e.g. full-time > part-time > unemployed > out of labor force); for multiple jobs, use the respondent-designated main job.

Harmonizing Employment Biographies: Pipeline



HILDA Employment Biography

- ▶ Built from `letter_wave_Combined.dta`: (i) a monthly employment calendar back to the start of the prior financial year, (ii) tenure variables (`letter_jbemlwk`, `letter_jjbemlyr`).
- ▶ Calendar allows up to 12 concurrent/sequential jobs; employment is prioritized over non-employment when sources overlap.
- ▶ Tenure at interview date helps infer job start dates before panel entry – extending coverage backward despite no full retrospective history.

SHP Employment Biography

- ▶ **Calendar:** `shpwave_p_user.dta`, monthly status between interviews; tenure variables `p_w66`, `p_w69` flag smooth job-to-job moves.
- ▶ **Retrospective:** `SHP0_bvwl_user.dta` (2001–2002) and `shpiii_prof_act_user.dta` (2013), yearly resolution, pre-panel-entry.
- ▶ On overlap, the calendar module wins (monthly timing, closer to interview, less recall bias).

PSID Employment Biography: Four Periods

- ▶ **1968–1976:** between-wave changes, household head only, no job-to-job moves.
- ▶ **1976–1987:** between-wave changes plus tenure measures; heads and spouses.
- ▶ **1988–2001:** employer-specific start/end dates for heads and spouses (no calendar file existed).
- ▶ **2003–2021:** monthly activity calendar combined with employer start/end dates; heads, spouses, and individual-level detail.

Caveat: biennial collection since 2003 raises recall bias; the household reference person answers for others, risking spousal misreporting; non-head/non-spouse members are observed only at interview.

Fertility Biography Example of Data

Example (wide format, one row per person)

pid	NR_KIDS	KID_1	KID_Y1	KID_M1	KID_S1	KID_2	KID_Y2	KID_M2	KID_S2	...
101	1	1	2017	2	2	0	NA	NA	NA	...
102	2	1	2012	5	1	1	2015	-1	2	...
103	0	0	NA	NA	NA	0	NA	NA	NA	...

Variable definitions

Variable	Meaning
pid	ID of the respondent
KID_1\$	Indicator: was there a child of order 1? (0/1)
KID_Y1\$	Year of birth of child 1 (1900–2024)
KID_M1\$	Month of birth of child 1 (–1 unknown, NA no child)
KID_S1\$	Child's biological sex (1 male, 2 female)
NR_KIDS	Total number of children in LIB

Partnership Variables: Example of Data

Example (wide format, one row per person)

pid	Spell identity						How the spell ended						PARTNERSHIP_ STATUS_2	..
	PARTNER_ ID_1	PARTNERSHIP_ STATUS_1	START_ Y_1	START_ M_1	END_ Y_1	END_ M_1	END_ SINGLE_1	END_ UNION_1	DIVORCE_ 1	DIVORCE_ Y_1	DIVORCE_ M_1			
101	5501	1	2014	11	2019	3	NA	1	0	NA	NA	2	..	
102	7702	2	2008	6	2016	1	NA	2	1	2016	9	0	..	
103	NA	0	2005	1	2020	4	1	NA	0	NA	NA	1	..	

Variable definitions

Spell identity

PARTNER_ID_\$	ID of respondent's partner
PARTNERSHIP_STATUS_\$	0 no union, 1 cohabitation, 2 marriage, 3 gap
START_Y/M_\$	Year / month the status started
END_Y/M_\$	Year / month the status ended

How the spell ended

END_SINGLE_\$	0 ongoing, 1 cohabitation, 2 marriage
END_UNION_\$	0 ongoing, 1 marriage, 2 separation, 3 death
DIVORCE_\$	Did a divorce occur? (0/1)
DIVORCE_Y/M_\$	Year / month of divorce

\$ = spell number: a union type with a specific partner. A cohabiting couple that marries becomes two spells (person 101: END UNION 1 = 1, marriage).

Employment Biography Example of Data

Example (wide format, one row per person)

pid	EMPLOYMENT_ STATUS_1	ENTRY_ Y1	ENTRY_ M1	EXIT_ Y1	EXIT_ M1	EMPLOYMENT_ STATUS_2	ENTRY_ Y2	EXIT_ Y2	...
101	1	2011	9	2016	10	3	2016	2017	...
102	3	2015	1	2018	6	1	2018	NA	...
103	1	2019	3	NA	NA	NA	NA	NA	...

Variable definitions

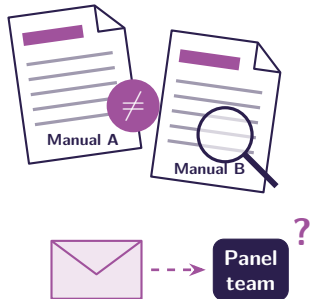
Variable	Meaning
EMPLOYMENT_STATUS_\$	1 working, 2 unemployed, 3 not in LF, 4 retired, 5 other
ENTRY_Y\$ / ENTRY_M\$ / ENTRY_D\$	Year / month / day entering the status
EXIT_Y\$ / EXIT_M\$ / EXIT_D\$	Year / month / day exiting the status

As with partnership, \$ indexes the spell number within a person's employment history.

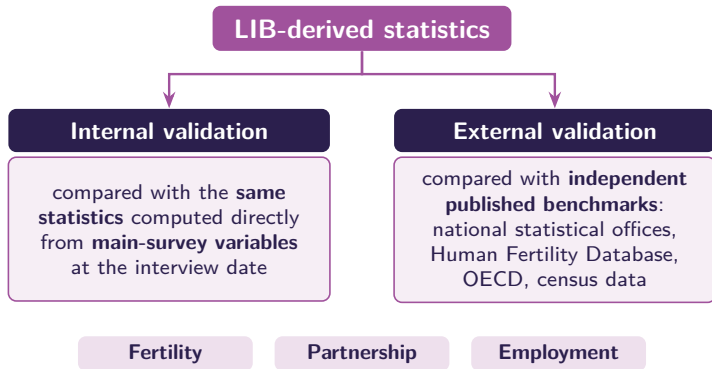
Validation of LIB

The hardest part of this project: finding the right sample restrictions

- ▶ Each panel combines several samples: refreshments, boosts, immigrant samples and non-sample members.
- ▶ We had to decide which of them to keep so that LIB matches external benchmarks.
- ▶ This required **reading panel manuals that sometimes contradicted each other.**
- ▶ It also meant **contacting the panel teams directly** to clarify which samples to keep.



Validation Approach



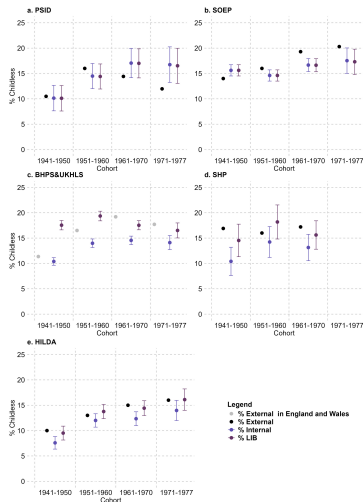
Seven figures cover all three domains;
this section walks through what each one shows.

Selecting Subsamples in Practice

- ▶ **SOEP refreshment samples:** ten labeled subsamples added since 1984 (A through R) – e.g. A = 1984 Initial Sample (West), C = 1990 Initial Sample (East), R = 2022 Refreshment. `psample` identifies which subsample a respondent belongs to.
- ▶ **BHPS/UKHLS:** `memorig` flags respondents from the ethnic-minority boost sample, so analyses can include or exclude it.
- ▶ **PSID:** `sample` distinguishes sample types (e.g. “Immigrant or Latino”, “SEO”, “SRC”); `drop` and `followable` flag 1997 dropout and follow-up eligibility.

These variables let you keep subsamples separate, pool them, or benchmark a refreshment sample against the original one – all through `01_data_selector.R`, without touching the underlying scripts.

Fertility Validation: Fig. 1. % Childless Women by Cohort



All panels: women younger than 40 in the year of their last interview are excluded

► PSID:

- Excluded: non-sample members, the moved-in sample, followable non-sample parents, and non-sample elderly.
- Restricted to families not dropped in 1997: dropped families may lack complete fertility histories
- Excluded: Immigrant Samples (1997, 1999, 2017, 2019) and Latino Samples (1990, 1992).

► SOEP: included samples A 1984 Initial (West), B 1984 Migration (until 1983, West), C 1990 Initial (East), D 1994/95 Migration (1984–1994, West), and Refreshments E 1998, F 2000, H 2006, J 2011, K 2012, R 2022.

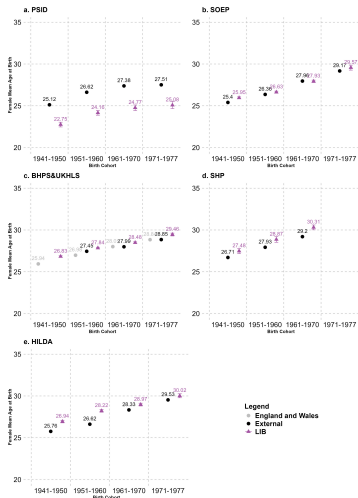
► BHPS & UKHLS: excluded the ethnic minority boost.

► SHP: excluded individuals absent from the retrospective file.

► HILDA: no additional restrictions.

► External benchmarks (% childless women by cohort): USA – Human Fertility Database (2024); Germany – Statistisches Bundesamt (2024); UK – Office for National Statistics (2024); Switzerland – census (2000); Australia – census (2021).

Fertility Validation: Fig 2. Female cohort mean age at birth



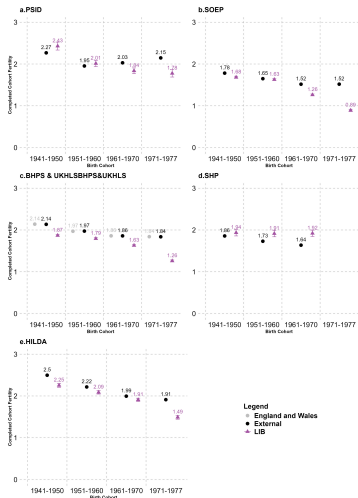
PSID:

- Excluded: non-sample members, the moved-in sample, followable non-sample parents, and non-sample elderly.
- Restricted to families not dropped in 1997: dropped families may lack complete fertility histories.
- Excluded: Immigrant Samples (1997, 1999, 2017, 2019) and Latino Samples (1990, 1992).

SOEP: included samples A 1984 Initial (West), B 1984 Migration (until 1983, West), C 1990 Initial (East), D 1994/95 Migration (1984–1994, West), and Refreshments E 1998, F 2000, H 2006, J 2011, K 2012, R 2022.

- BHPS & UKHLS:** excluded the ethnic minority boost.
- SHP:** excluded individuals absent from the retrospective file.
- HILDA:** no additional restrictions.
- External benchmarks** External Female cohort mean age at birth for all birth orders - Human Fertility Database (2024)

Fertility Validation: Fig 3. Completed Cohort Fertility at 40 by cohort



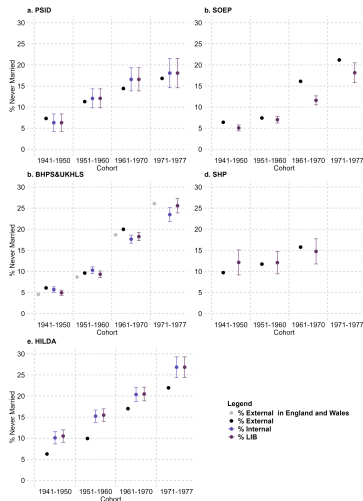
PSID:

- Excluded: non-sample members, the moved-in sample, followable non-sample parents, and non-sample elderly.
- Restricted to families not dropped in 1997: dropped families may lack complete fertility histories.
- Excluded: Immigrant Samples (1997, 1999, 2017, 2019) and Latino Samples (1990, 1992).

SOEP: included samples A 1984 Initial (West), B 1984 Migration (until 1983, West), C 1990 Initial (East), D 1994/95 Migration (1984–1994, West), and Refreshments E 1998, F 2000, H 2006, J 2011, K 2012, R 2022.

- BHPS & UKHLS:** excluded the ethnic minority boost.
- SHP:** excluded individuals absent from the retrospective file.
- HILDA:** no additional restrictions.
- External benchmarks** USA, Germany, United Kingdom, Switzerland - Human Fertility Database (2024), Australia (Human Collection Database, 2025)

Partnership: Fig 4. % Never Married by Cohort



► PSID:

- Excluded: non-sample members, the moved-in sample, followable non-sample parents, and non-sample elderly.
- Restricted to families not dropped in 1997: dropped families may lack complete fertility histories.
- Excluded: Immigrant Samples (1997, 1999, 2017, 2019) and Latino Samples (1990, 1992).

- **SOEP:** included samples A 1984 Initial (West), B 1984 Migration (until 1983, West), C 1990 Initial (East), D 1994/95 Migration (1984–1994, West), and Refreshments E 1998, F 2000, H 2006, J 2011, K 2012, R 2022.

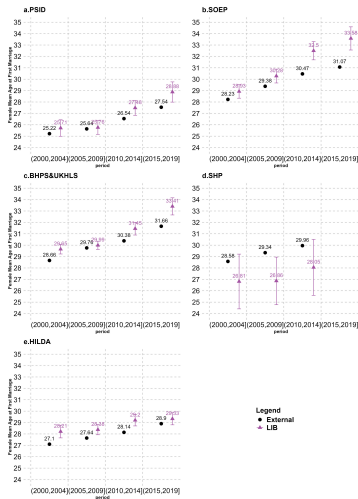
- **BHPS & UKHLS:** excluded the ethnic minority boost

- **SHP:** included only individuals present in the retrospective files.

- **HILDA:** no additional restrictions.

- **External: Percent Never Married Women by Cohort** United Kingdom - ONS(2024) USA, Germany , Switzerland, Australia - (United Nations [UN], 2019)

Partnership: Fig 5. Mean Age at First Marriage



► PSID:

- Excluded: non-sample members, the moved-in sample, followable non-sample parents, and non-sample elderly.
- Restricted to families not dropped in 1997: dropped families may lack complete fertility histories.
- Excluded: Immigrant Samples (1997, 1999, 2017, 2019) and Latino Samples (1990, 1992).

► SOEP: included samples A 1984 Initial (West), B 1984 Migration (until 1983, West), C 1990 Initial (East), D 1994/95 Migration (1984–1994, West), and Refreshments E 1998, F 2000, H 2006, J 2011, K 2012, R 2022.

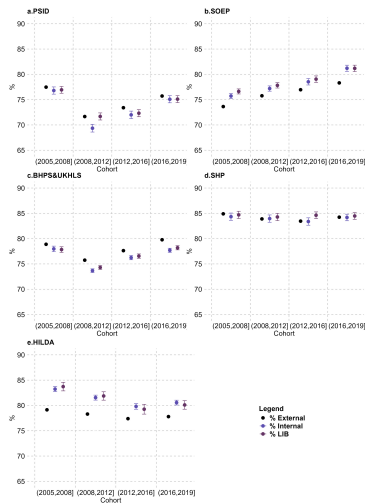
► BHPS & UKHLS: excluded the ethnic minority boost

► SHP: included only individuals present in the retrospective files.

► HILDA: no additional restrictions.

► External Period Female Mean Age at First Marriage - OECD (2025)

Employment: Fig 6. Male Labor Force Participation



All panels: only individuals aged 15 to 64 are included.

► PSID:

- Excluded: non-sample members, the moved-in sample, followable non-sample parents, and non-sample elderly.
- Restricted to families not dropped in 1997: dropped families may lack complete fertility histories.
- Excluded: Immigrant Samples (1997, 1999, 2017, 2019) and Latino Samples (1990, 1992).

► SOEP:

included samples A 1984 Initial (West), B 1984 Migration (until 1983, West), C 1990 Initial (East), D 1994/95 Migration (1984–1994, West), and Refreshments E 1998, F 2000, H 2006, J 2011, K 2012, R 2022.

► BHPS & UKHLS:

excluded the ethnic minority boost.

► SHP:

no additional restrictions.

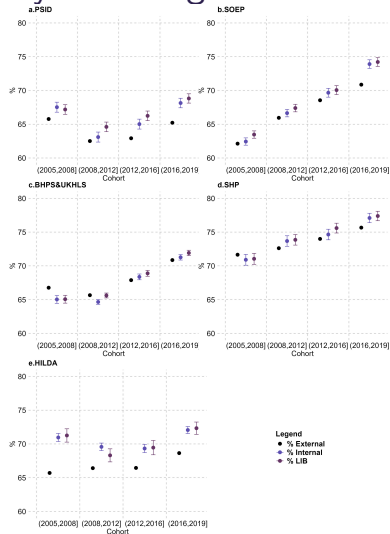
► HILDA:

no additional restrictions.

► External benchmark:

male employment – OECD (2025).

Employment: Fig. 7. Female Labor Force Participation



► PSID:

- Excluded: non-sample members, the moved-in sample, followable non-sample parents, and non-sample elderly.
- Restricted to families not dropped in 1997: dropped families may lack complete fertility histories.
- Excluded: Immigrant Samples (1997, 1999, 2017, 2019) and Latino Samples (1990, 1992).

► SOEP: included samples A 1984 Initial (West), B 1984 Migration (until 1983, West), C 1990 Initial (East), D 1994/95 Migration (1984–1994, West), and Refreshments E 1998, F 2000, H 2006, J 2011, K 2012, R 2022.

► BHPS & UKHLS: excluded the ethnic minority boost.

► SHP: no additional restrictions.

► HILDA: no additional restrictions.

► External benchmark: female employment – OECD (2025).

All panels: only individuals aged 15 to 64 are included.

What the Validation Tells Us, Overall



Reliable

Across **fertility, partnership and employment**, LIB faithfully represents the underlying life-course processes.



Minor deviations

Confined to specific cohorts or panels, most often the **oldest and youngest PSID cohorts**.



Robust & comparable

The pattern holds across **five very different survey designs**.

Take-away: LIB measures are robust and comparable across countries

PSID

SOEP

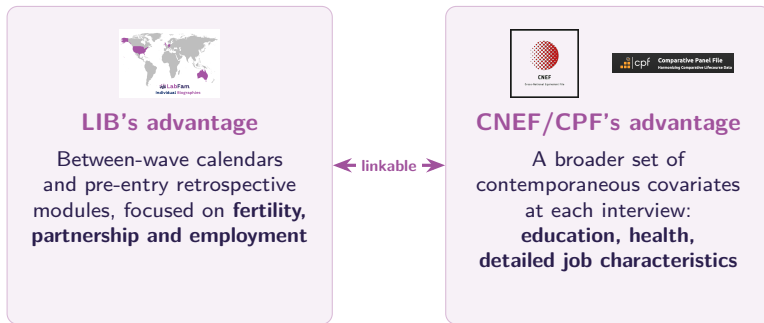
BHPS/UKHLS

SHP

HILDA

Compatibility with CNEF/CPF

LIB is designed to **complement, not replace**, **CNEF** (Frick et al. 2007) and **CPF** (Turek, Kalmijn, & Leopold 2021).



Linked via **respondent identifiers** and **country codes**

Merging LIB with the Comparative Panel File

1 LIB biographies

long format: one row per person-month

pid	cc	month	status	kids	union
101	UK	01/2016	FT	0	cohab.
101	UK	02/2016	FT	0	cohab.
101	UK	03/2016	Inactive	1	cohab.
101	UK	04/2016	Inactive	1	cohab.
...					

2 CPF



one row per person-interview

you need to run the CPF code first

pid	cc	int.	health	educ
101	UK	02/2016	good	tertiary
101	UK	02/2017	fair	tertiary
...				



3

merge on pid, country, interview date

pid	cc	month	status	kids	union	health	educ
101	UK	01/2016	FT	0	cohab.	NA	NA
101	UK	02/2016	FT	0	cohab.	good	tertiary
101	UK	03/2016	Inactive	1	cohab.	NA	NA
101	UK	04/2016	Inactive	1	cohab.	NA	NA
...							

NA: CPF values attach only at interview months; all other months are **missing by construction** and, depending on the research question, may call for imputation.

CPF is distributed as code, not data: build it with the CPF scripts (Turek, Kalmijn, & Leopold 2021), then merge. The merge code is included among the 28 LIB scripts.

Running Code

Getting Started: What You Need to Run LIB

1

Install R (the engine) from cran.r-project.org. LIB was written and tested in **R 4.4.0**; use this version or newer.

2

Install RStudio Desktop (the dashboard: editor, console, data viewer) from posit.co. Both are free.

3

Download the LIB code from [GitHub](#) (*Code* → *Download ZIP*), unzip it and open `lib.Rproj`.

4

Open `00_setting_work_space.R`, set `folder_personal` to the folder with your panel data (Stata `.dta` files), and run it: it creates the folders and installs and loads all **23 packages** LIB needs.

Working with the LIB Codebase - in RStudio



How to install it?<https://www.youtube.com/watch?v=TFGY1KvQEQ4>

Packages in R: The Stata Analogy

A **package** (also called a *library*) is an add-on that brings new commands, just like a user-written ado command in Stata.

	Stata	R
Install once	<code>ssc install estout</code>	<code>install.packages("dplyr")</code>
Use in each session	automatic after installing	<code>library(dplyr)</code> every session
Where add-ons live	SSC, net install	CRAN (plus GitHub)
Script file	do-file (<code>.do</code>)	R script (<code>.R</code>)
Project folder	cd to a directory	RStudio project (<code>.Rproj</code>)
Reading .dta	<code>use file.dta</code>	<code>haven::read_dta("file.dta")</code>

Good news: `00_setting_work_space.R` does both steps for you: it installs any missing package and loads all of them. You only need an internet connection the first time.

The R Packages Behind LIB

Data wrangling

filter, reshape and join

dplyr

tidyr

data.table

collapse

purrr

stringr

reshape2

Reading panel data

Stata files and their labels

haven

labelled

sjlabelled

readr

openxlsx

Dates and spells

expand spells into months

lubridate

neatRanges

PSID helpers

PSID-specific tools

easyPSID

psidR

Validation graphs

figures and survival curves

ggplot2

patchwork

survival

survminer

Utilities

missing values, helpers

nanianr

sjmisc

rlang

Code Overview

- ▶ 28 R scripts, written in R version 4.4.0 (open source), plus a readme and a validation suite.
- ▶ Modular and configurable: users select country, biography, time period, and output format.
- ▶ Files are self-contained by design and should not be modified, so outputs stay identical across users.

Two Files Every User Should Know

- ▶ `00_setting_work_space.R` – all file paths and required packages for the rest of the pipeline.
- ▶ `01_data_selector.R` – which biographies, from which panels, get built (default: all biographies, all panels); also sets export format.
- ▶ Three export formats available: `.csv`, `.dta`, `.Rdata`.

Full Script Inventory

#	Database	Script(s)	Role
0	–	000_readme_file_LIB.txt	Project readme
1	–	00_setting_work_space.R	Paths & packages
2	–	01_data_selector.R	Run configuration
3	–	02_CPF_merging.R	Link to CPF

Country pipelines: four scripts each (master file, fertility, partnership, employment)

4–7	BHPS/UKHLS	four-file pattern	UK pipeline
8–11	SOEP	four-file pattern	Germany pipeline
12–15	HILDA	four-file pattern	Australia pipeline
16–20	PSID	four files + PSID_variables_needed_21_century.R	USA pipeline
21–24	SHP	four-file pattern	Switzerland pipeline

Validation: the scripts behind every figure in the Technical Validation section

25	all	validation_fertility.R	Fertility checks
26	all	validation_employment.R	Employment checks
27	all	validation_partnership.R	Partnership checks
28	all	validation_graph_functions.R	Plotting helpers

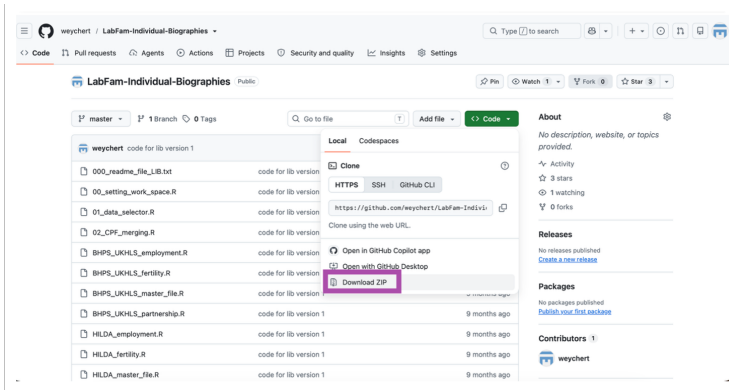
Go to <https://zenodo.org/records/22902836>
Download all codes and unzip them.



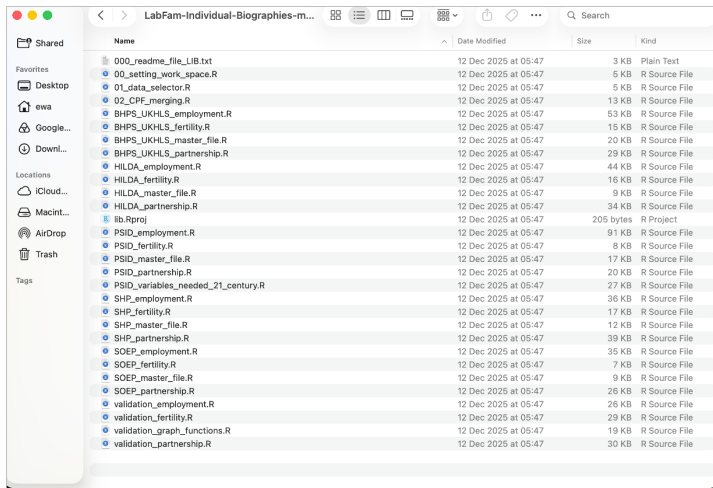
...or get the code from GitHub

Go to <https://github.com/weychert/LabFam-Individual-Biographies>

Download all codes and unzip them.



The unzipped folder should look like this



Name	Date Modified	Size	Kind
000_readme_file_LIB.txt	12 Dec 2025 at 05:47	3 KB	Plain Text
00_setting_work_space.R	12 Dec 2025 at 05:47	5 KB	R Source File
01_data_selector.R	12 Dec 2025 at 05:47	5 KB	R Source File
02_CPF_merging.R	12 Dec 2025 at 05:47	13 KB	R Source File
BHPS_UKHLS_employment.R	12 Dec 2025 at 05:47	53 KB	R Source File
BHPS_UKHLS_fertility.R	12 Dec 2025 at 05:47	15 KB	R Source File
BHPS_UKHLS_master_file.R	12 Dec 2025 at 05:47	20 KB	R Source File
BHPS_UKHLS_partnership.R	12 Dec 2025 at 05:47	29 KB	R Source File
HILDA_employment.R	12 Dec 2025 at 05:47	44 KB	R Source File
HILDA_fertility.R	12 Dec 2025 at 05:47	16 KB	R Source File
HILDA_master_file.R	12 Dec 2025 at 05:47	9 KB	R Source File
HILDA_partnership.R	12 Dec 2025 at 05:47	34 KB	R Source File
lib.Rproj	12 Dec 2025 at 05:47	205 bytes	R Project
PSID_employment.R	12 Dec 2025 at 05:47	91 KB	R Source File
PSID_fertility.R	12 Dec 2025 at 05:47	8 KB	R Source File
PSID_master_file.R	12 Dec 2025 at 05:47	17 KB	R Source File
PSID_partnership.R	12 Dec 2025 at 05:47	20 KB	R Source File
PSID_variables_needed_21_century.R	12 Dec 2025 at 05:47	27 KB	R Source File
SHP_employment.R	12 Dec 2025 at 05:47	36 KB	R Source File
SHP_fertility.R	12 Dec 2025 at 05:47	17 KB	R Source File
SHP_master_file.R	12 Dec 2025 at 05:47	12 KB	R Source File
SHP_partnership.R	12 Dec 2025 at 05:47	39 KB	R Source File
SOEP_employment.R	12 Dec 2025 at 05:47	35 KB	R Source File
SOEP_fertility.R	12 Dec 2025 at 05:47	7 KB	R Source File
SOEP_master_file.R	12 Dec 2025 at 05:47	9 KB	R Source File
SOEP_partnership.R	12 Dec 2025 at 05:47	26 KB	R Source File
validation_employment.R	12 Dec 2025 at 05:47	26 KB	R Source File
validation_fertility.R	12 Dec 2025 at 05:47	29 KB	R Source File
validation_graph_functions.R	12 Dec 2025 at 05:47	19 KB	R Source File
validation_partnership.R	12 Dec 2025 at 05:47	30 KB	R Source File

Step 2: Download data from the data holders

HILDA

The screenshot shows the HILDA website with a navigation bar at the top. The main content area is titled 'Data access' and contains information about how to access the data. It mentions that users need to register with the UK Data Service and that there are different levels of access (e.g., summary statistics, individual data). The text is small and difficult to read, but the structure is clear.

<https://melbourneinstitute.unimelb.edu.au/hilda/for-data-users>

SOEP

The screenshot shows the SOEP website with a navigation bar at the top. The main content area is titled 'Data access' and contains information about how to access the data. It mentions that users need to register with the SOEP Research Data Center (SDC) and that there are different levels of access (e.g., summary statistics, individual data). The text is small and difficult to read, but the structure is clear.

https://www.diw.de/en/diw_01.c.601584.en/data_access.html

BHPS / UKHLS

Access Understanding Society data

The screenshot shows the Understanding Society website with a navigation bar at the top. The main content area is titled 'Data access' and contains information about how to access the data. It mentions that users need to register with the UK Data Service and that there are different levels of access (e.g., summary statistics, individual data). The text is small and difficult to read, but the structure is clear.

<https://www.understandingsociety.ac.uk/documentation/access-data/>

PSID

The screenshot shows the PSID website with a navigation bar at the top. The main content area is titled 'Data access' and contains information about how to access the data. It mentions that users need to register with the PSID Research Center and that there are different levels of access (e.g., summary statistics, individual data). The text is small and difficult to read, but the structure is clear.

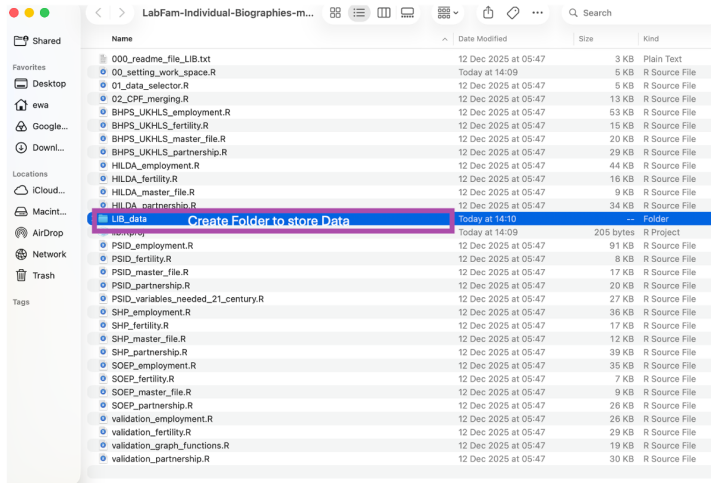
<https://simba.isr.umich.edu/DC/f.aspx>

SHP

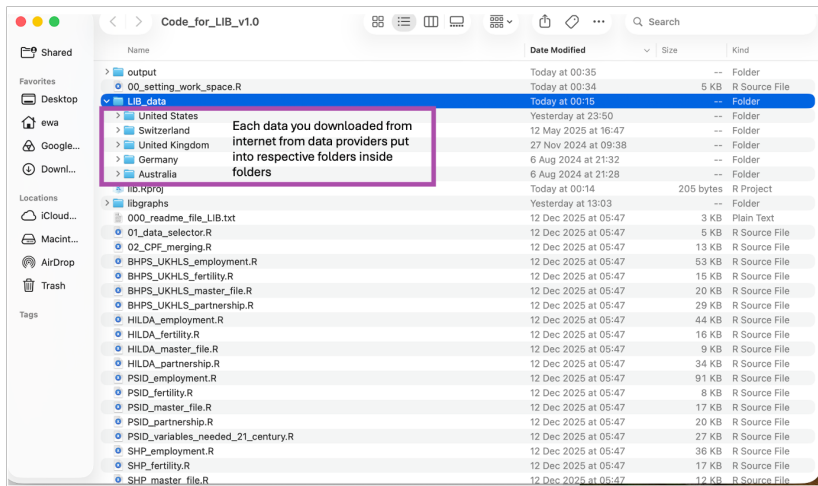
The screenshot shows the SHP website with a navigation bar at the top. The main content area is titled 'Data access' and contains information about how to access the data. It mentions that users need to register with the SHP Research Center and that there are different levels of access (e.g., summary statistics, individual data). The text is small and difficult to read, but the structure is clear.

<https://forscenter.ch/projects/swiss-household-panel/data/>

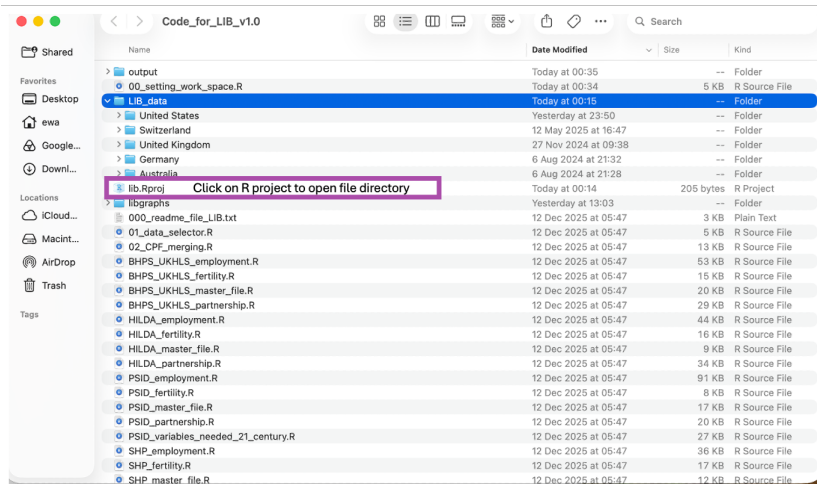
Step 3: Create a folder to store the data



Put each downloaded dataset into its country folder



Step 4: Open the R project



Set the working directory to your own path

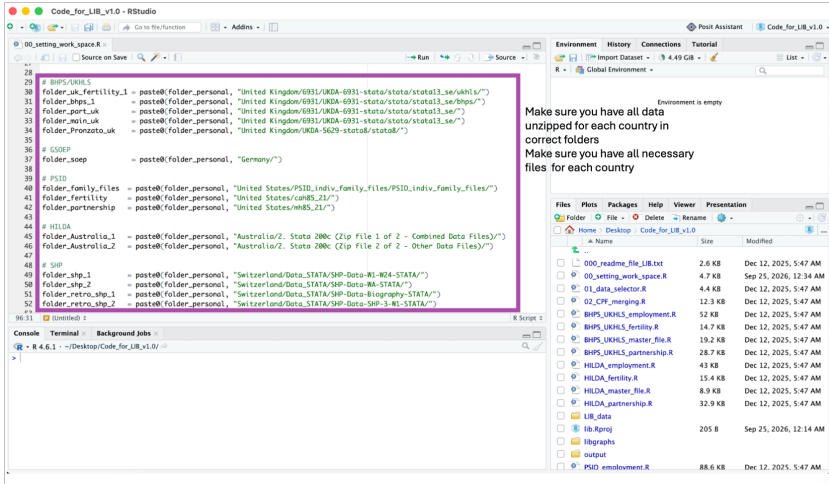
The screenshot displays the RStudio environment with the following components:

- Source Editor:** Contains R code for setting the working directory. A red box highlights the line: `folder_personal = "/Users/ewa/Desktop/Code_for_LIB_v1.0/LIB_data/"`. A text annotation next to it reads: "Change working directory of the project to your own".
- Environment Panel:** Shows "Environment is empty". A purple box overlay reads: "Where data are store in current R session".
- Files Panel:** Shows a list of files in the current working directory. A purple box overlay reads: "Current working directory files shown".
- Console:** At the bottom, with a purple box overlay reading: "Console – where results will be shown".

Files Panel Content:

File Name	Size	Modified
000_readme_file_LIB.txt	2.6 KB	Dec 12, 2025, 5:47 AM
00_setting_work_space.R	4.7 KB	Sep 25, 2026, 12:34 AM
01_data_selector.R	4.4 KB	Dec 12, 2025, 5:47 AM
02_CPF_merging.R	12.3 KB	Dec 12, 2025, 5:47 AM
BHPS_UKHLS_employment.R	52 KB	Dec 12, 2025, 5:47 AM
BHPS_UKHLS_fertility.R	14.7 KB	Dec 12, 2025, 5:47 AM
BHPS_UKHLS_master_file.R	19.2 KB	Dec 12, 2025, 5:47 AM
BHPS_UKHLS_partnership.R	28.7 KB	Dec 12, 2025, 5:47 AM
HILDA_employment.R	43 KB	Dec 12, 2025, 5:47 AM
HILDA_fertility.R	15.4 KB	Dec 12, 2025, 5:47 AM
HILDA_master_file.R	8.9 KB	Dec 12, 2025, 5:47 AM
HILDA_partnership.R	32.9 KB	Dec 12, 2025, 5:47 AM
lib_data		
lib.Rproj	205 B	Sep 25, 2026, 12:14 AM
libgraphs		
output		
PSID_employment.R	88.6 KB	Dec 12, 2025, 5:47 AM

Check the data paths for each country



Code_for_LIB_v1.0 - RStudio

00_setting_work_space.R

```
28 # BHPS/UKHLS
29 folder_uk_fertility_1 = paste0(folder_personal, "United Kingdom/6931/UKDA-6931-stata/stata13_se/ukhls/")
30 folder_bhps_1 = paste0(folder_personal, "United Kingdom/6931/UKDA-6931-stata/stata13_se/bhps/")
31 folder_part_uk = paste0(folder_personal, "United Kingdom/6931/UKDA-6931-stata/stata13_se/")
32 folder_main_uk = paste0(folder_personal, "United Kingdom/6931/UKDA-6931-stata/stata13_se/")
33 folder_Pronzato_uk = paste0(folder_personal, "United Kingdom/UKDA-5629-stata8/stata8/")
34
35 # GSOEP
36 folder_soep = paste0(folder_personal, "Germany/")
37
38 # PSID
39 folder_family_files = paste0(folder_personal, "United States/PSID_indiv_family_files/PSID_indiv_family_files/")
40 folder_fertility = paste0(folder_personal, "United States/cah85_21/")
41 folder_partnership = paste0(folder_personal, "United States/mh85_21/")
42
43 # HILDA
44 folder_Australia_1 = paste0(folder_personal, "Australia/2. Stata 200c (Zip file 1 of 2 - Combined Data Files)/")
45 folder_Australia_2 = paste0(folder_personal, "Australia/2. Stata 200c (Zip file 2 of 2 - Other Data Files)/")
46
47 # SHP
48 folder_shp_1 = paste0(folder_personal, "Switzerland/Data_STATA/SHP-Data-W1-W24-STATA/")
49 folder_shp_2 = paste0(folder_personal, "Switzerland/Data_STATA/SHP-Data-WA-STATA/")
50 folder_retro_shp_1 = paste0(folder_personal, "Switzerland/Data_STATA/SHP-Data-Biography-STATA/")
51 folder_retro_shp_2 = paste0(folder_personal, "Switzerland/Data_STATA/SHP-Data-SHP-3-W1-STATA/")
```

Environment is empty

Make sure you have all data unzipped for each country in correct folders
Make sure you have all necessary files for each country

Files | Plots | Packages | Help | Viewer | Presentation

Name	Size	Modified
000_readme_file_LIB.txt	2.6 KB	Dec 12, 2025, 5:47 AM
00_setting_work_space.R	4.7 KB	Sep 25, 2026, 12:34 AM
01_data_selector.R	4.4 KB	Dec 12, 2025, 5:47 AM
02_CPF_merging.R	12.3 KB	Dec 12, 2025, 5:47 AM
BHPS_UKHLS_employment.R	52 KB	Dec 12, 2025, 5:47 AM
BHPS_UKHLS_fertility.R	14.7 KB	Dec 12, 2025, 5:47 AM
BHPS_UKHLS_master_file.R	19.2 KB	Dec 12, 2025, 5:47 AM
BHPS_UKHLS_partnership.R	28.7 KB	Dec 12, 2025, 5:47 AM
HILDA_employment.R	43 KB	Dec 12, 2025, 5:47 AM
HILDA_fertility.R	15.4 KB	Dec 12, 2025, 5:47 AM
HILDA_master_file.R	8.9 KB	Dec 12, 2025, 5:47 AM
HILDA_partnership.R	32.9 KB	Dec 12, 2025, 5:47 AM
LIB_data		
lib.Rproj	205 B	Sep 25, 2026, 12:14 AM
libgraphs		
output		
PSID_employment.R	88.6 KB	Dec 12, 2025, 5:47 AM

Step 5: Open 01_data_selector.R – choose panels

The screenshot shows the RStudio interface with the file `01_data_selector.R` open in the editor. The script contains comments and R code for setting up the environment and selecting data panels. A purple box highlights the `country_set` variable, and a text overlay indicates that the user should choose which panels they want to run.

```
1 #-----#
2 # Labfam Individual Biographies #
3 #-----#
4
5 # This is the main editable file for the LIB. Users choose countries and histories.
6 # Based on these choices, the program calls the necessary scripts from the database.
7 # The default is to build all histories for all countries.
8
9 # List of required packages for this task and establish working directory
10 source("00_setting_work_space.R")
11
12 country_set = toupper(c("hilda", "soep", "bhps_ukhls", "shp", "psid"))
13
14
15 # Available histories: "employment", "fertility", "partnership"
16 history_set = c("master", "fertility", "partnership", "employment")
17
18 # Preferred format for the output database: "csv", "rds", "dta", etc.
19 type_you_want = ".dta"
20
21 #####
22
23 # End of customization part. Subsequent lines call all the databases and
24 # export the data
25
26 #####
27
28 # Create object with all desired histories to iterate
29 path <- "output"
```

Choose which panels you want to run

The Environment pane on the right shows the Global Environment with an empty list of objects. The Files pane at the bottom right shows the directory structure of the project, including various R scripts and data files.

Open 01_data_selector.R – choose histories

The screenshot shows the RStudio interface with the `01_data_selector.R` script open. The script contains comments and code for selecting countries and histories. A purple box highlights the line `history_set = c("master", "fertility", "partnership", "employment")`, with the text "Choose which histories you want to run" overlaid. The file explorer on the right shows the project structure, including files like `01_data_selector.R`, `02_CPE_merging.R`, and various history files.

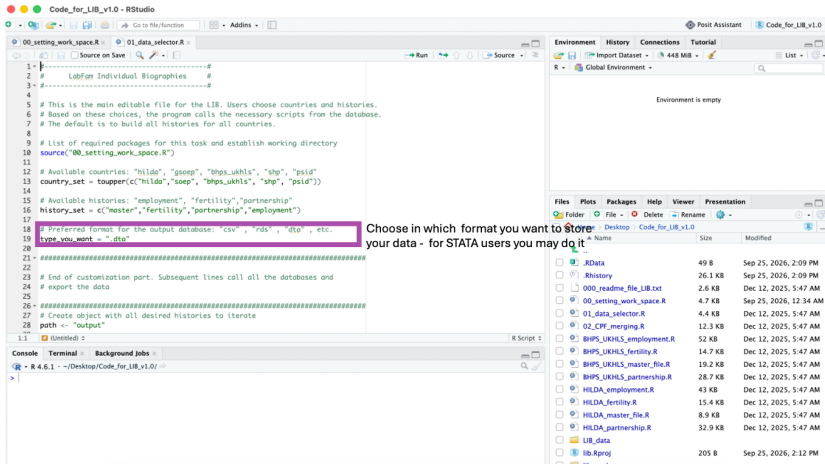
```
1 #-----#
2 # Labfam Individual Biographies #
3 #-----#
4
5 # This is the main editable file for the LIB. Users choose countries and histories.
6 # Based on these choices, the program calls the necessary scripts from the database.
7 # The default is to build all histories for all countries.
8
9 # List of required packages for this task and establish working directory
10 source("00_setting_work_space.R")
11
12 # Available countries: "hilda", "gsoep", "bhps_ukhls", "shp", "psid"
13 country_set = toupper(c("hilda", "soep", "bhps_ukhls", "shp", "psid"))
14
15 #-----#
16 history_set = c("master", "fertility", "partnership", "employment")
17 #-----#
18 # Preferred format for the output database: "csv", "rds", "dta", etc.
19 type_you_want = ".dta"
20
21 #-----#
22
23 # End of customization part. Subsequent lines call all the databases and
24 # export the data
25
26 #-----#
27 # Create object with all desired histories to iterate
28 path <- "output"
```

Environment: Global Environment

Files: Home Desktop Code_for_LIB_v1.0

Name	Size	Modified
..		
..RData	49 B	Sep 25, 2026, 2:09 PM
..Rhistory	26.1 KB	Sep 25, 2026, 2:09 PM
000_readme_file_LIB.txt	2.6 KB	Dec 12, 2025, 5:47 AM
00_setting_work_space.R	4.7 KB	Sep 25, 2026, 12:34 AM
01_data_selector.R	4.4 KB	Dec 12, 2025, 5:47 AM
02_CPE_merging.R	12.3 KB	Dec 12, 2025, 5:47 AM
BHPS_UKHLS_employment.R	52 KB	Dec 12, 2025, 5:47 AM
BHPS_UKHLS_fertility.R	14.7 KB	Dec 12, 2025, 5:47 AM
BHPS_UKHLS_master_file.R	19.2 KB	Dec 12, 2025, 5:47 AM
BHPS_UKHLS_partnership.R	28.7 KB	Dec 12, 2025, 5:47 AM
HILDA_employment.R	43 KB	Dec 12, 2025, 5:47 AM
HILDA_fertility.R	15.4 KB	Dec 12, 2025, 5:47 AM
HILDA_master_file.R	8.9 KB	Dec 12, 2025, 5:47 AM
HILDA_partnership.R	32.9 KB	Dec 12, 2025, 5:47 AM
LIB_data		
lib.Rproj	205 B	Sep 25, 2026, 2:12 PM

Open 01_data_selector.R – choose output format



The screenshot shows the RStudio interface with the script `01_data_selector.R` open. The script contains comments and code for setting up the environment and selecting output formats. A purple box highlights the following lines in the script:

```
18 # Preferred format for the output database: 'csv', 'rds', 'dtp', etc.  
19 type_you_want = ".dtp"
```

Overlaid on the right side of the script is the text: "Choose in which format you want to store your data - for STATA users you may do it".

The Files pane on the right shows a list of files in the `Code_for_LiB_v1.0` directory. The files and their sizes are:

Name	Size	Modified
.RDdata	49 B	Sep 25, 2026, 2:09 PM
.Rhistory	26.1 KB	Sep 25, 2026, 2:09 PM
000_readme_file_LiB.txt	2.6 KB	Dec 12, 2025, 5:47 AM
00_setting_work_space.R	4.7 KB	Sep 25, 2026, 12:34 AM
01_data_selector.R	4.4 KB	Dec 12, 2025, 5:47 AM
02_CPF_merging.R	12.3 KB	Dec 12, 2025, 5:47 AM
BHPS_UKHLS_employment.R	52 KB	Dec 12, 2025, 5:47 AM
BHPS_UKHLS_fertility.R	14.7 KB	Dec 12, 2025, 5:47 AM
BHPS_UKHLS_master_file.R	19.2 KB	Dec 12, 2025, 5:47 AM
BHPS_UKHLS_partnership.R	28.7 KB	Dec 12, 2025, 5:47 AM
HILDA_employment.R	43 KB	Dec 12, 2025, 5:47 AM
HILDA_fertility.R	15.4 KB	Dec 12, 2025, 5:47 AM
HILDA_master_file.R	8.9 KB	Dec 12, 2025, 5:47 AM
HILDA_partnership.R	32.9 KB	Dec 12, 2025, 5:47 AM
LiB_data		
lib.Rproj	205 B	Sep 25, 2026, 2:12 PM

Step 6: Select all lines and run the script

1. I choose bhps ukhls fertility in .dta format

2. I marked all lines of code (cmd + A) and

3. click Run at the top right-hand corner of the script

4. Wait and enter folder output in 3 minutes you should see data on bhps ukhls in .dta format

Step 7: Run the script

The screenshot shows the RStudio interface with the 'bhaps_ukhls_fertility' dataset loaded into the Environment pane. The dataset has 150,279 observations and 66 variables. The console shows the script execution output, including the workspace loading and the successful loading of the dataset from the RDS file.

Environment Pane: bhaps_ukhls_fert_ 150279 obs. of 66 variables

Console Output:

```
[Workspace loaded from ~/Desktop/Code_for_LIB_v1.0/RData]
> bhaps_ukhls_fertility <- readRDS("~/Desktop/Code_for_LIB_v1.0/output/bhaps_ukhls_fertility.rds")
> View(bhaps_ukhls_fertility)
>
```

Files Pane:

Name	Size	Modified
~		
bhaps_ukhls_fertility.dta	39 MB	Sep 25, 2026, 2:18 PM
bhaps_ukhls_fertility.rds	1.1 MB	Sep 25, 2026, 2:18 PM
LIB_2026_09_25.dta	80.8 MB	Sep 25, 2026, 2:18 PM
master_bhaps_ukhls_wide.rds	4 MB	Sep 25, 2026, 2:18 PM

Inspect the output in STATA

Data Editor (Edit) — bhps_ukhls_fertility.dta

pidp[1] 2727

	pidp	NR_KIDS	KID_Y1	KID_M1	KID_S1	KID_1	KID_Y2	KID_M2
1	1	1966	11	female	child	.	.	.
2	1	1981	6	male	child	.	.	.
3	1	1979	8	female	child	.	.	.
4	2	1966	9	female	child	1972	6	mc
5	2	1966	9	female	child	1972	6	mc
6	2	1980	9	male	child	1988	3	mc
7	2	1980	9	male	child	1988	3	mc
8	3	1993	6	female	child	1995	5	fer
9	1	1995	2	female	child	.	.	.
10	1	1988	10	female	child	.	.	.
11	2	1988	10	female	child	2002	2	fer
12	3	1986	3	male	child	1988	5	mc
13	1	2003	12	male	child	.	.	.
14	2	1987	2	female	child	1997	6	fer
15	2	2016	2	male	child	2019	1	fer
16	2	1992	3	male	child	1994	11	mc
17	1	2005	10	female	child	.	.	.
18	2	2009	12	male	child	2011	6	mc
19	2	1946	7	male	child	1948	8	fer
20	2	1944	7	female	child	1948	8	mc
21	4	1964	9	female	child	1968	11	fer
22	3	1968	11	female	child	1971	2	fer
23	1	2003	11	female	child	.	.	.
24	2	1964	5	male	child	1965	6	mc
25	3	1980	10	male	child	1982	2	fer
26	3	1980	10	male	child	1982	2	fer
27	7	1957	4	female	child	1958	5	fer
28	3	1986	3	male	child	1988	5	mc
29	2	1946	12	female	child	1950	12	fer
30	2	1946	12	female	child	1950	12	fer

Vars: 66 Order: Dataset Obs: 150,279

You may open and see data in STATA

Variables

Name	Label
pidp	
NR_KIDS	Number of c...
KID_Y1	year of birth...
KID_M1	month of bir...
KID_S1	sex of 1 child
KID_1	birth order...
KID_Y2	year of birth...
KID_M2	month of bir...
KID_S2	sex of 2 child
KID_2	birth order...
KID_Y3	year of birth...
KID_M3	month of bir...
KID_S3	sex of 3 child
KID_3	birth order...
KID_Y4	year of birth...

Properties

Variables

Name	Label
pidp	
Type	double
Format	%10.0g
Value label	
Notes	

Data

Frame	Label
default	
File name	bhps_ukhls_fertility.dta
Label	
Metafile	

Filter: Off



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LabFam Individual Biographies harmonised family and employment histories based on panel surveys

Ewa Weychert, Beata Osiewalska, Lucas van der Velde, Anna Matysiak

This is a preprint; it has not been peer reviewed by a journal.

<https://doi.org/10.21203/rs.3.rs-8376548/v1>
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Abstract

Reproducibility in social science is often hindered by inconsistent data preparation and limited transparency. The LabFam Individual Biographies (LIB) project addresses this challenge by providing open, cross-national harmonization of life-course histories from five long-running panels: Australia (HILDA), Germany (SOEP), Switzerland (SHP), the United Kingdom (BHPS/UKHLS), and the United States (PSID). LIB reconstructs spell-based data across three domains—fertility (number and timing of births), partnership (timing of union formation/dissolution), and employment (employment spells and job characteristics)—by integrating panel questionnaires, calendar modules, and retrospective components to recover events between waves and before survey entry. Outputs are organized as dated spells with explicit starts and ends, enabling duration and transition analyses within and across countries. We document variable definitions, harmonization and conflict-resolution procedures, and validation against internal survey indicators and external benchmarks. Implemented in R and released as open code, LIB supports complete reproduction, user customization, and linkage to complementary infrastructures (e.g., CPF/CNEF), thereby reducing setup costs and improving comparability.

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<https://doi.org/10.21203/rs.3.rs-8376548/v1>

Where to Go Next

- ▶ **Code at GitHub:** github.com/weychert/LabFam-Individual-Biographies
- ▶ **Code at Zenodo:** zenodo.org/records/22902836
- ▶ **Data note preprint:** <https://doi.org/10.21203/rs.3.rs-8376548/v1>
- ▶ **Contact:** e.weychert@uw.edu.pl

Research Using LIB

When Politics Enters the Family: Electoral Outcomes and Fertility in the United Kingdom

Baszczak, Ł., & Weychert, E. (2026). Working Paper No. 2026-17.

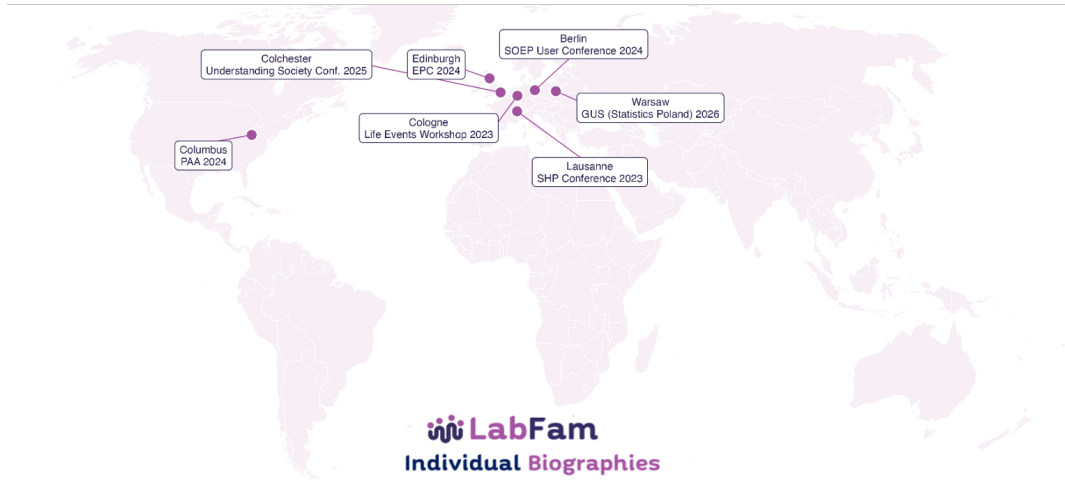
Climate Change Narratives and First Births in the UK

Weychert, E., Vignoli, D., Matysiak, A., & Celińska-Kopczyńska, D. (2026). Working Paper No. 2026-11.

Work and Childbearing in the XXI Century: When Large Work Autonomy Meets High Job Demands

Matysiak, A., Osiewalska, B., & Weychert, E.

LIB at conferences





LabFam

LabFer



Ewa Weychert

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